

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	KS THARUNESHWAR	Reg. No.:	192572148
Section / Batch:	2025	Date:	27-08-26

Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identifies all key issues.
- Thorough analysis with validated results and performance evaluation

Question Type	Description	Questions	Total Marks
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1 – Q2	100

SECTION A – Industry Problem Solving Task

**SIMATS Online Programming Lab**

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DAA PROGRAMMING

CO4-AT2-Industry problem solving

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Language: Python C C++ Java

QUESTIONS

Q1
Greedy Technique
50 marks

Q2
Greedy Technique
50 marks

2/2 answered

Q1 Greedy Technique 50 marks

Cloud Task Scheduling Optimization (Greedy + Priority Scheduling)
A cloud platform must schedule multiple computational tasks efficiently across virtual machines. Analyze how greedy algorithms and priority queues can optimize scheduling. Identify constraints such as task dependencies and resource limits. Design a scalable scheduling solution. Discuss implementation challenges. Evaluate time complexity and system performance. Interpret results in distributed computing environments.

Your Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 typedef struct {
5     int task_id;
6     int priority;
7     int execution_time;
8 } Task;
9
10 void swapTasks(Task *a, Task *b) {
11     Task temp = *a;
12     *a = *b;
13     *b = temp;
14 }
15 void sortTasksGreedy(Task tasks[], int n) {
16     for (int i = 0; i < n - 1; i++) {
17         for (int j = 0; j < n - i - 1; j++) {
18             if (tasks[j].priority < tasks[j + 1].priority ||
19                 (tasks[j].priority == tasks[j + 1].priority &&
20                  tasks[j].execution_time > tasks[j + 1].execution_time))
21                 swapTasks(&tasks[j], &tasks[j + 1]);
22         }
23     }
24 }
```

Input
stdin input

Output
Execution Order:
Task 104 (Priority: 5, Exec Time: 1 min) -> Start: 0
min, End: 1 min
Task 102 (Priority: 5, Exec Time: 2 min) -> Start: 1

CO4-AT2-Industry problem solving

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Language: Python C C++ Java

QUESTIONS

Q1

Greedy Technique
50 marks

Q2

Greedy Technique
50 marks

2/2 answered

Q2 Greedy Technique

50 marks

AI-Based Resume Screening Optimization (Search + Ranking)

A recruitment platform must screen thousands of resumes to identify suitable candidates. Analyze how search and ranking algorithms can optimize candidate selection. Identify constraints such as keyword matching and skill relevance. Design an efficient filtering system. Discuss scalability in large hiring platforms. Evaluate computational complexity. Interpret results in improving hiring efficiency.

Your Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4
5 typedef struct {
6     int candidate_id;
7     char name[50];
8     int matched_keywords;
9     float skill_relevance_score;
10 } Candidate;
11
12 void swapCandidates(Candidate *a, Candidate *b) {
13     Candidate temp = *a;
14     *a = *b;
15     *b = temp;
16 }
17 void rankCandidatesGreedy(Candidate candidates[], int n) {
18     for (int i = 0; i < n - 1; i++) {
19         for (int j = 0; j < n - i - 1; j++) {
```

Input

stdin input

Output

```
--- AI-Based Resume
Screening Optimization ---
Ranked Shortlist (Top
Candidates First):
Rank 1 | ID: 3 | Name:
Charlie | Matched
```

Code of Conduct

I V Deepa Dharshini (192525213) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: KS
THARUNESHWAR

RUBRICS FOR CALCULATION

Case Study	Understanding of Industry Problem (20)	Application of Engineering Concepts (25)	Solution Design (25)	Use of Tools (15)	Analyses (10)	Professionalism (5)	Total Marks (Each - 50)
Q1							
Q2							
	Overall Total (100)						
Faculty In-charge		Course Coordinator			Head of Department		
Name: _____		Signature: _____			Signature: _____		
Signature: _____							

Date: _____	Date: _____	Date: _____
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Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Mark Evaluation: